MONICA NURSING STUDY CENTER

Cardiovascular

Study Guide

Organized review material

Prepared for study and printing

Cardiovascular

- **Ischemic Cardia Disease**
  1. Angina
  2. MI

Angina

Types of Angina

- Stable angina (chronic or exertional)
- Unstable angina
- Printzmetal variante or vasospastic)

Stable Angina (chronic or exertional)

- Chronic or exertional
- Occur with activities, involve exertion or emotional like having sex (elderly), stress.
- Stable pattern: 3 months or more, Onset, duration, relieving factor.
- Relief with rest and nitroglycerin.

Unstable Angina

- Also call pre-infarction angina.
- Occur with unpredictable physical or emoctional of exertion or emotion.
- Pain During rest
- Any change in stable pattern

Printzmetal or vasospastic

- Pain during night, often awake clients with pain
- Attack may be associated with EKG changes
  - ST Segment Elevation.
  - Negative T.

Esta se trata with CCB, Diltiazem, verapamil nifedipine = (el Raynaud disease se trata igual)

Assessment Angina

1. Chest pain mild to moderate substernal, crushing, squeezing, may radiate to left side: Arm, jaw, neck, back.
   - Is unaffected by inspiration or expiration.
   - Duration less than 5 minutes (Relieve with nitro or rest usually)
2. SOB, pallor, sweating, palpitation, tachycardia, dizziness, and faintness.
3. Digestive Disturbance.

Note: Epigastric Pain

- MI
- Pre-eclampsia eclampsia
- Pancreatitis
- Addisonian crisis
- Diabetes ketoacidosis

Diagnostic Studies

1. EKG normal except in the case of Printzmetal or vasospastic:
   - ST segment elevation
   - T Negative

   Ischemia
2. Stress test- Chest pain
3. Cardiac enzymes
   - CK-MB.

   Normales

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- Troponin Negative

4. Cardiac catheterization. = (Diagnostic and Treatment)

Diagnostic and Treatment

1. Safety position: sit the Pt
2. Administer 1st Nitroglycerine sublingual,

5 min after first nitro if not relieve call 911, and administer 2nd dose of nitro,
5 min after second administer 3rd Nitro. =
(3 in 15 min period with 5 min apart between them)
Note Nitro: photosensitivity = contained Ambar, thigh close, dura 90 dias after that discard bc se vence.

- ✓ Da headache and its expective, just give Tylenol.
- ✓ Da orthostatic hypotension
- ✓ No with Viagra (sildenafil), grapefruit.
- ✓ Burning or tingling sensation under tongue is expective

Following Acute Episode

1. Risk factor modification
   - -Exercises, diet, Low Na, low fat
     - No alcohol, No smoking. K= OK (salt substitute)
2. Aspirin o clopidogrel
3. Anticoagulants
4. Statin (decreased cholesterol)

Myocardial Infraction

- Occur when myocardial Tissue is abruptly or severe deprive of O2.

Assessment

Same to angina

- Pain is more severe and not relieved with nitroglycerine.

➢ **Atypical manifestation**

- ✓ Women: Fatigue, epigastric pain, SOB, NONSTEMI (no ST segment elevation).
- ✓ Elderly: Confusion, LOC alteration, CHF symptoms, SOB
- ✓ Diabetes: Without pain

Location of MI

- Left descending anterior artery = Anterior or septal wall or septal MI.
- Circumflex artery= posterior or lateral MI
- Right coronary artery= Inferior wall MI

Diagnostic MI

1. EKG
   - ✓ -ST segment elevation
   - ✓ - Negative T
   - ✓ - Pathologic Q= Necrosis
2. Cardiac enzymes
   a. **CK-MB** – Begin to rise 3-4 hours after 72 hours. Lasting elevated 72 hours. Board: if after 72 hours lasting elevate o al 4to dia = re-infarction.
   b. **Myoglobin**- Level rises 2-3 hours after, lasting elevated 6 hours.
   c. **Troponin-** More specific (global test)
   Beginning to rise 3-4 hours after, MI lasting elevated 7-10 days.(same to amylase and lipase in pancreatitis)
   d. LDH level Elevated

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e. **WBC (5000-10000)**= 20000-25000 (is normal here) same in pregnant during labor and newborn.

3. Stress test
4. Thallium scam
5. Cardia catheterization = diagnostic and treatment

Complications of cardias catheterization.

- Hemorrhage
- Decrease or absent pf peripheral pulse
- MI = bc haciendote el procedimiento se le puede causar otro MI

Intervention

1. Hospital admission
2. MONA

M O N A	Morphine O2 Nitroglycerine Aspirin (chewable)	Board: viejita con MI no quiere ir al hospital. Le doy aspirina.

3. Main TX

Thrombolytic therapy, within 6 hours

- ✓ Alteplase
- ✓ Tenecteplase
- ✓ Recombinant streptoquinase

Side effect

Hemorrhage: Hemorrhagic Stroke = CT scan before

If client have:

- Slurred speech
- LOC alteration, Confussion
- Paralysis

Stop Thrombolytic therapy and Call the Dr.

4. Medications

- Aspirin and clopidogrel
- Betablockers and prils to protect myocardium

5. Coronary stent
6. Coronary graft
7. PTCA angioplasty = percutaneous transluminal coronary angioplasty = same to cardiac catheterization =check peripheral pulses
8. Risk factor modification

Following Acute Episode

1. Bed rest for 24-36 hours
2. Progress to dangling legs at the side of the bed to the chair for 30 minutes 3 times per day.
3. Progress to ambulation.
4. Allow the client to stand to void or use bedside commode.
5. Sexual intercourse is allowed when a client is able to climb 2 flight of stairs without SOB.

MI Complication

1. Heart failure
2. Cardiac dysrhythmias
   - PVCS
   - 1 or 2 PVCs after MI = Priority= Call HCP
   - V Taq

- V Fib
- Asystole

3. Cardiogenic shock
4. Pericarditis
5. Cardiac tamponade
6. Post-infarction Angina
7. Dressler's Syndrome:
   - ✓ Pericarditis
   - ✓ Pericardial effusion
   - ✓ Pleural effusion

Heart Failure
Inability of the heart to maintain adequate cardiac output to meet the metabolic needs of the body.

Right Side CHF	Left side CHF
- Dependent Edema ♦ Leg or sacral - Jugular vein distended (explore on <45) - Abdominal distension - Hepatomegaly - Splenomegaly - Weight gain - Anorexia, nausea and weight gain. - Nocturnal diuresis - Ascites - Generalized edema. Note: weigh gain = 2 pounds per day 5 per weeks	- Signs of pulmonary congestion - SOB - Tachypnea - Crackles - Paroxysmal nocturnal, Dyspnea - orthopnea - Dry, hacking cough Empeorando needs more pillows Mejorando menos pillows

Complication of CHF

1. Acute pulmonary edema = life threatening emergency.
   - ✓ Expectoration of large amount of pink blood = **pink frothy sputum = hemoptysis.**

Note: Hematemesis = GI bleeding and is bright red

- ✓ Ascending crackles more than ½ half of both lungs = le esta subiendo la sangre.
- ✓ Severe SOB,
- ✓ Severe Orthopnea
- ✓ Profuse sweeting
- ✓ Cyanosis
- ✓ Nasal flaring.
- ✓ Pallor
- ✓ Tachycardia
- ✓ Anxiety
- ✓ Apprehension
- ✓ Bubbling respiration

Diagnostic (CHF)
BNP = (Type B Natriuretic peptide)
>100 = CHF

Intervention in Acute setting

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1. Place client in Hight fowler position 90 = sitting = same in autonomic dysreflexia
2. O2
3. IV line for medication, No Fluids.

Diuretics = Furosemide (choice) in high dose
Morphine = bc anxiety y dye sensation
Digoxin (tto elección para insuficiencia cardiaca
Niseritide (only use in uncompensated CHF) hijo del BNP

Following Acute episode (tto Casa)

1. Medications
   - Digoxin
   - ACE INHIBITORS (PRILES)= (cardio and renal protectors)
   - Low dose of B-blockers
2. Risk factor modification
3. No OTC medications
4. Avoid caffeine, tea, coca-cola, chocolate
5. Cardiac Diet= low fat, low Na, low cholesterol, fluid restriction if uncompensated

Diet high K to prevent digoxin toxicity (allow salt substitute)

6. Avoid Isometric Exercises.

2. Pericarditis

- Acute or chronic inflammation of the pericardium that cause compression, loss of the pericardial elasticity and accumulation of fluid

Assessment

1. Chest pain same to angina or MI with the difference that is Grating and aggravated by breathing (particularly inspiration, coughing and swallowing

- Precordial in anterior chest that radiated to the left side of the neck, shoulder or back.
- Worsening in supine position

Board: nurse put a client in supine position= negligence

- May be relieved by learning forward position.

2. Pericardial Friction Rub.

- Scratchy high-pitched sound

3. Fever and chills in case of infection.
4. Fatigue
5. Malaises
6. Increased WBC
7. EKG change: ST Segment Elevation, Atrial fibrillation

Intervention

1. Position client in high fowler or leaning forward.
2. Medication:

- ✓ Analgesic
- ✓ NSAID = Ibuprofen, celecoxib = no with sulfa allergy
- ✓ Colchicine = Acute gout (inflammation)
- ✓ Steroids
- ✓ Antibiotic in case of bacterial infection
- ✓ Diuretic
- ✓ Digoxin

- ✓ In the case of chronic pericarditis: **Pericardiectomy**.

Cardiac Tamponade

- Pericardial effusion occurs when the space between the parietal and visceral layer of the pericardium fills up with fluid.

Assessment

1. Beck's Triad
    - Venous hypertension (JVD) with clear lungs. (if dirty is another thing)
    - Hypotension arterial
    - Muffled or distant heart sounds.
2. Pulse paradoxes, (same in asthmatic status)
    - Decreased systolic BP more than 10mmhg at the end of inspiration.
3. Widening pulse pressure
4. Sign of decreased cardiac output

Intervention

1. ICU Admission
- Hemodynamic Monitoring
2. IV Fluid
3. Prepared for pericardiocentesis (bc fluid)
- In case of recurrent= Pericardial window or pericardiectomy if recurrent or chronic.

3. Cardiogenic shock (heart no working)

- It is a failure of the heart to pump adequately reducing cardiac output.
- Compromise tissue perfusion

Assessment

1. Hypotension +tachycardia= Shock)
    - BP less than 90/60mmhg (or below base line).
2. Decreased urinary output, less than 30ml/hr
3. Confusion, LOC alteration, restlessness
4. Cold, clammy skin, sweeting
5. Pulmonary Congestion sign
6. Tachypnea
7. Chest discomfort.

Intervention

1. Morphine
2. O2
3. Prepare for intubation and mechanical ventilation
4. Diuretics
5. Vasopressor
    - Dopamine
    - Dobutamine shock
    - Epinephrine
6. Assist with the insertion of swan-ganz catheter

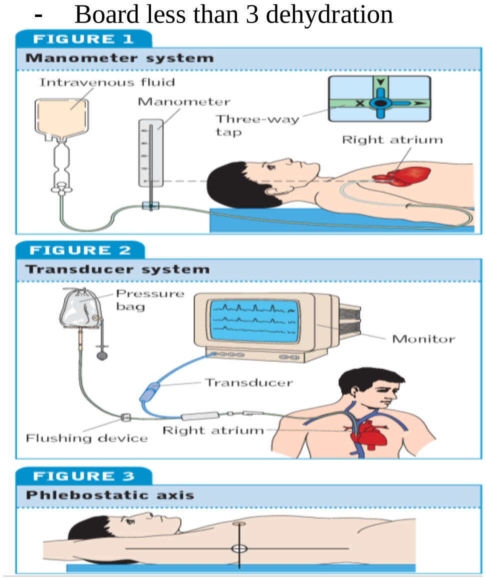
Entra por el right atrium hasta la arteria pulmonal
Have 3 lumen

1. Distal port to check PCWP (normal 4-12) (Pulmonary artery pressure)
    - PCWP NORMAL =4-12
    - >12= Fluid overload
    - < 4 = Fluid deficit volume
2. Proximal port to measure CVP (normal 3-7)

- ➢ CVP Measurement

- The client needs to be supine with HOB-45. (Same in jugular vein distention)

- Zero point on the transducer needs to be at the level of right atrium= phlebostatic Axis.
- The right atrium localized at:
  - ✓ The midaxillary line
  - ✓ The 4 intercostal spaces
- Need to be relaxed, avoid coughing or straining.
- More than 7= fluid overload
- Board less than 3 dehydration



4. Endocarditis

- Inflammation of the inner lining of the heart and valves.

Causes:

Occur primarily who are:

1. IV drug Abuser
2. Rheumatic fever

- Had have valve replacement

Port of entry

- Oral cavity, especially client has had a dental procedure in the previous 3-6 months.
- 1-2 weeks before dental procedure poner prophylactic antibiotics

Assessment

1. Fever
2. Anorexia
3. Weight loss
4. Fatigue
5. Cardiac Murmur
6. Clubbing Fingers (no priority bc lung term)
7. Heart Failure
8. Petechias
9. Splinter hemorrhage in the nail beds
10. Janeway lession
11. Osler's Nodes

This last 3 are painfull, red rise lession found on hads and fingers

12. Embolic complications

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Embolic complication (septic embolism) (Endocarditis complication)

- Stroke
  - Slurred Speech, paralysis, weakness.
- Pulmonary Embolism
  - SOB, tachycardia, cyanosis hemoptysis
- Renal Embolism
  - Flank pain + hematuria
- Splenic Embolism
  - Board like abdominal pain radiate to left shoulder + hypovolemic shock

Intervention

1. Monitor for complications
2. Antibiotic- more than 2 weeks (Penicillin) PICC line (central line in place + than 6 weeks)
   - ✓ 4-6 weeks
   - ✓ Note: **mid-line** short term version of PICC use for 2-3 weeks
3. Homecare Instruction
   - ✓ Aseptic technique during IV antibiotic administration.
   - ✓ Record temperature daily for up to 6 weeks and report fever
   - ✓ Encourage oral Hygiene at least twice daily with soft toothbrush and rinse well after.
   - ✓ Avoid use oral irrigation device and flossing
   - ✓ Administered prophylactic antibiotic 1-2 weeks before invasive procedure.

Valve replacement procedure

Tx for rheumatic fever or any cardiac valve disease

1. Mechanical prosthetic valve

Mechanical or plastic

- ✓ More risk for thrombus = anticoagulant lifelong
- ✓ More risk for infection
- ✓ More duration (30years)

2. Bioprosthetic valve

Using biological graft

- ✗ Xenograft (pig or cow)
- ✗ Homograft (human cadaver)
- ✓ More risk for thrombus
- ✓ Anticoagulants for 3-6 months
- ✓ Lasting 10-45

Normal heart sound

- ➢ SI – Closure of tricuspid and mitral valve heard best at the 5th intercostal space.
- ➢ S2 Closure of Aortic and pulmonary valve best heard in the 2nd left intercostal space.
- ➢ S3 Ventricular gallop hear in the Apex may be normal in children and pregnant. Adult=CHF.
- ➢ S4 Always pathologic = report always
  - ✓ Ischemic or hypotension (call Dr)
  - ✓ Ventricular Hypertrophy
- Aortic area
  - – 2nd intercostal space just to the right sternum.
- Pulmonic Area

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- 2nd intercostal space just to the left to the sternum.
- Tricuspid area
  - 5th Intercostal space at the lower left of sternum border.
- Mitral
  - 5th Intercostal space at the apex 5th ICS Mid clavicular line CLM (left).

Vascular Disorders

1. **Venous**
- **Phlebitis**

-Vein inflammation
-Redness and Hot

- **Thrombophlebitis**

-Superficial venous thrombosis
-Cord formation or streak through vein wall
-Redness and Hot

2. **Arterials**
- DVT

Intervention for both phlebitis and thrombophlebitis

- ✓ Stop IV line immediately
- ✓ Remove IV line and use the other hand.
- ✓ Apply warn moist compress
- ✓ NO SE ELEVA EL BRASO

Deep Vein Thrombosis (DVT)

Cause or risk factors

- Immobility
- Long Travels (board)
- Pregnant
- After surgery bc immobility
- Obesity
- Oral contraceptive

Assessment

1. Unilateral Calf or groin tenderness or pain with or without swelling
2. Homan's sign positive
   - ✓ Calf pain when **dorsiflex foot**
3. Warm skin, redness

Complications

Pulmonary embolism

- ✓ SOB
- ✓ Tachypnea
- ✓ Cyanosis
- ✓ Chest pain
- ✓ Hemoptisis

Intervention DVT

1. Bed rest for at least 72 hours with elevated legs, above heart level.

Early ambulation is preventive

2. Avoid knee-Gatch, never pillow under knee

3. Do not massage
4. Provide tight-high or knee- high Antiembolism stocking to increase venous return.
5. Administer warm moist compress
6. Administered thrombolytic therapy (5-7 days)
    - Alteplase
    - Tenecteplase
7. Anticoagulants Heparin to prevent cloth formation, warfarin at home
8. Diuretics

Note: Thrombolytic therapy
DVT = 5-7 days
Stroke = 4-5 hours
MI = 6 hours

Venous insufficiency

- Result from prolonged venous hypertension which stretches the vein and damages the valves.

Assessment

1. Stasis dermatitis or brown discoloration along the ankles extending to the calf.
2. Edema.
3. Ulcer formation:
    - -At the ankle level
    - -Uneven edges
    - - Moist
    - - Pink base
    - - Granulation tissue
    - - Better prognosis than Arterial.

Intervention

1. Elevated the extremities to increase venous return for 10-20 minutes every few hours each day.
2. Instruct the client to wear elastic or compression stocking during the day and evening. (Remove before going to bed, put on awake before getting out the bed).
3. Avoid prolonged sitting or standing, avoid constrictive clothes or crossing legs
4. Use intermittent sequential compression device (SCDs) Twice daily for 1 hour.

Varicose veins

- Distended, protruding vein that appear darkened and tortuous

Assessment

- ✓ Pain in the legs with dual aching (fatigue or cansansio en la pantorrilla) after standing.
- ✓ Ankle edema.
- ✓ Felling of fullness sensation in the legs

Diagnostic

- Trendelenburg test
- ✓ Place client in a supine position with elevated legs
- ✓ Veins fill from proximal to distal in case of varicose veins
- ✓ Normally fill from distal to proximal when are elevated.

Note

Trendelenburg sign = Developmental hip dysplasia.
Trendelenburg position = Variable deceleration in OB, risk for preterm labor, Aneurism, (bc se puede romper), hypotension, air embolism.

Intervention

- Same to venous insufficiency

Peripheral Arterial Disease

Arterial insufficiency

- ✓ Chronic disorder in which partial or total arterial occlusion deprives the Lower Extremities of O2 and nutrients.

Cause

- Atherosclerosis is the most common cause.

Assessment

1. Intermittent claudication
   - ✓ Pain in the muscle resulting from inadequate blood supply.
   - ✓ Instruct the client to walk to the point of claudication, stop and rest, and then walk a little farther.
2. Rest pain (polyneuropathic) characterized by:
   - Numbness, burning pain, paresthesia in the distal portion of the lower extremities which awakens the client at night and
   - Relieve by place the extremities dangling on dependent position.
3. Lower back and buttock discomfort.
4. Loss of hair, dry scaly skin in L/E.
5. Thickened toenails.
6. Cold and gray-blue color
7. Decreased or absent peripheral pulse.
8. Ulcer formation
   - ✓ On or between the toes or on the upper aspect of the foot.
   - ✓ Painful
   - ✓ No granulation tissues.
   - ✓ Dry, pales base
   - ✓ Worse prognosis
   - ✓ Necrosis + amputation is frequent.

Intervention

1. Assess for ulcer formation or gangrene
2. Affected. Limb dangling, or depended position
3. Exercise to develop collateral circulation.
4. Avoid alcohol, caffeine, tobacco.
5. Avoid apply direct heat, (no heating pad to prevent injury)
6. Medication
   - Pentoxifylline.
   - Antiplatelet (clopidogrel),
   - ASA
7. Surgical intervention
   - Angioplasty = cortar la artery los pedasos malos y empatarla
   - Atherectomy = remover la placa de atheroma
   - Bypass surgery = stein

   Note:
   Carotid: son las que llevan la sangre para el brain
   Coronary artery: llevan la sangre para el corazon
   Phemoral artery: para las piernas

Raynaud's disease

- Vasospasm causes constriction of the cutaneous vessels.
- Attacks are intermittent and occur with exposure to cold and stress
- Affecting primarily finger, nose, toes, ears, and cheeks

Assessment

1. Red, white, and blue fingers
2. Blanching of the extremities followed by cyanosis during vasoconstriction and redness when vasospasm is relieved.
3. Paresthesia (numbness, tingling en los dedos)

Intervention.

1. Stop caffeine, stop smoking.
2. Vasodilators CCB
   - Diltiazem
   - Verapamil

   Antidysrhythmic
   - Nifedipine --- Antihypertensive.
3. Avoid Cold exposure
   - ✓ Use gloves and warm water for washing dishes.
   - ✓ Use glove when cold temperature.
   - ✓ Wear warm clothing and socks

Buerger's Disease – Thromboangiitis Obliterans.

- Is an occlusive disease of the median and small Arteries and Vein.
- More frequent in women's associate with smoking.
- Frequent amputations

Assessment

1. Intermittent claudication.
2. Ischemic pain occurs in the digits while at rest, more severe at night.
3. Paresthesia.
4. Extremities are cold.
5. Ulcer formation.
6. Necrosis and frequent amputation.

Intervention

- Stop Smoking
- Vasodilators
- CCB

Aortic Aneurism

- Abnormal dilation of the arterial wall, caused by localized weakness and stretching of an artery.
- DO NOT PALPATE

Thoracic	Abdominal
1. Pain extended to the neck and shoulder 2. Syncope 3. SOB 4. Tachycardia 5. Cyanosis 6. Weakness 7. Hoarseness and difficulty to swallowing	1. Prominent pulsating mas in abdomen at or above umbilicus 2. Systolic bruit over the Aorta. 3. Tenderness on Deep Palpation. (Do no palpate) Si ya tiene la Aneyrism, ya la masa esta, aqui se preoriza el lower back pain bc rupture.

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✓ Priority, call the Dr.	

Rupturing of Aneurism (Priority)	
1. Lower back pain 2. Sign of Hypovolemic shock ✓ Hypotension BP less than 90/60 ✓ Tachycardia ✓ Increased HR 3. Bluish periumbilical discoloration and Flank Hematoma = Internal bleeding.	Intervention 1. Administer BP medication. 2. Surgical Intervention = Aneurism resection 3. Rupturing = IV fluid Activate Code.

Anemias

Iron Deficiency Anemia

Causes

1. Decreased absorption
2. Intermittent bleeding
   - ✓ Ulcerative colitis
   - ✓ Crohn's disease
   - ✓ Esophageal varices
3. Chronic gastritis = decrease absorption of iron
4. Toddler milk Anemia
   - ✓ Decreased milk intake to 32 oz per day.
5. Pregnant
   - ✓ 1st trimester = Physiologic Anemia

Assessment

1. Fatigue, SOB, Tachycardia.
   - ✓ Activity intolerance.
2. PICA (eats unfood substances = grass, ice, powder)
3. Cheilitis and Cheilosis
   - Labios cortados. – Boquera.
4. Spoon nails.

Diagnostic Anemia

1. HB ↓
2. Hematocrit ↓
3. Transferrin ↑ (TIBC) (trasporte)
4. Serum Iron ↓
5. Ferrites ↓ (almacen de Iron)
6. Microcytic hypochromic Anemia.

Reticulocytes Count (baby hemoglobin)

- Early improvement 2 weeks (up).
- While HB improved 1 months after

Por lo tanto, cuando no han pasado 2 semanas se hace esta prueba.

Treatment

1. Energy conservation intervention.
2. Increased iron in Diet
   - Red Meat - Shellfish + Oranges or

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- Organ Meat - Dark green
- Lentils - eggs (yolk) Tomatoes juice
- Raisin - Tuna
- Legume to Fix the Iron

3. Iron supplemts
   - Sulfate, Fumarate ferrous, gluconato ferrous
   - ✓ With empty stomach
   - ✓ With straws to prevent teeth discoloration
   - ✓ With orange and tomatoes juice
   - ✓ Constipation and tarry stool are expected.
4. IM Iron Dextrin in Z Track Technique.
   - ✓ Do not massage.
   - ✓ Do not use deltoid method.

Megaloblastic Anemia

1. Folic Acid Deficiency:

Causes: Associate with

- ✓ Pregnant
- ✓ Alcohol
- ✓ Phenobarbital
- ✓ Sulfa

2. Pernicious Anemia= Vit B-12 Deficiency

Cause

1. Elderly = Chronic gastritis
2. Gastric surgery after 6 months:
   - ✓ Bypass Surgery
   - ✓ Bilrow 1
   - ✓ Bilrow 2
   - ✓ Bariatric Surgery
3. Vegans (12 months after)

Assessment

1. Red beefy Tongue

Note: TONGUES
Kawasaki = strawberry tongue
Scarlet fever = strawberry tongue
Furry tongue = Antibiotics S/E penicillin and cephalosporin
Red beefy tongue = Pernicious anemia

2. Fatigue, SOB, Activity Intolerance
3. Neurologic Symptoms
   - ✓ Paresthesias
   - ✓ Ataxia by impair gait.

Diagnostic

- ➢ Schilling Test
- × Urinary collection 24hrs (discard 1st am, start after that recollecting 24h)

To confirm diagnostic:
Less than 200 or 10 % of vit B12 in Urine = Pernicious Anemia

Treatment

1. Vegan Vit B supplement
   - Choice Tofu
2. Vit B 12 (Cyanocobalamin)
   - 1000 mg IM daily for 4 weeks
   - Then weekly IM until hematocrit = normal
   - Monthly lifelong IM.

Aplastic Anemia

Causes

- Radiation
- Chemotherapy
- Viral
- Toxins

Assessment

1. Pancytopenia

↓ WBC = Risk for Infection
↓ RBC = Anemia
↓ Platelets = Risk for bleeding

Diagnostic

- Bone marrow biopsy
- Se hace en el hip
- Bedside procedure
- Over affected side after

Treatment

1. Blood transfusion
2. ↑ HB = Epoetin ∝. (NO IN LEUKEMIA)
   - To increase RBC
   - SQ or IV x 3 weeks.
   - Do not shake.
   - Discard unused.
   - Start when HB less than 10.5.
   - And stop when HB in 12 and Hematocrit = 33 bc
3. Filgrastim (Neupogen) to ↑ WBC
   - SQ OR IV
   - Continued until Neutrophil count = 10 000

Side effects

- Bone and Join Pain
- Loss of Hair
- ↑Uric Acid and Cholesterol
- Leukocytosis

4. Oprelvekin = To increased Platelets S/E = CHF
5. Neutropenic and bleeding precautions

Sickle Cell Anemia

- Hemoglobinopathies in which HB A is replaced by HB S.
- Autosomal recessive (25% if both parent carriers)
- African American (life expect around 50 years)

Board: Pediatric Nurse Floating give him/her this client.

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Never Never Diuretics
Administered Pneumococcal Vaccine (bc splenectomy)

- Trigger Factor of Vaso-occlusive Crisis
  - Dehydration
  - Fever
  - Emotional and physical stress
  - Infection
  - High Altitud
  - Surgery
  - Blood loss
  - COLD WATER

    (They can swim only in warm water) Never in a cool pool.

Assessment

1. Hand- Foot Syndrome (1st Symptoms of SCA
   - Bone infarctions cause Painful, Swelling of hand and feet.
2. Vaso – Occlusive Crisis = Thrombosis
   - Causa by stasis of blood = ischemia = IM
   - Painful swelling hand, feet, Abdominal pain, back pain
   - Priapism
3. Splenic Sequestration and Aplastic Crisis.
   - Severe Anemia
   - Hypovolemic shock
   - Jaundice bc destruction.

Complication

- ✓ Hearth failure
- ✓ Pulmonary infarction
- ✓ Retinal hemorrhage
- ✓ Renal damage
- ✓ Stroke
- ✓ Osteoporosis
- ✓ Pneumonia
- ✓ Shok

Intervention

H- Hydration
O - O2
P - Pain medication
- Morphine = PCA pump (check LOC before)

Note: Here morphine is used, no meperidine to prevent seizure.
It's the opposite to Cholecystitis no morphine use meperidine instead.

2. Diet
- ✓ High Calories
- ✓ High Protein
- ✓ Folic Acid Supplement

Note: blood transfusion at bed side.

3. Ensure that the child received Pneumococcal vaccine.

   Client with splenectomy must have the pneumococcal vaccine 1st.
4. Splenectomy = client recurrent splenic sequestration.
5. Avoid cold temperature.

6. Hydroxyurea (Anti-cancer) preventive to prevent Vaso occlusive crisis
    - ◊ Risk for infection (teratogenic)

Note: Used in polycythemia vera to prevent thrombosis.

Hemophilia

- Hereditary Bleeding disorder.
- X- linked (mom carrier – son sick)

Group of bleeding disorder resulting from deficiency of specific coagulation factor

- Hemophilia A = Factor VIII 8 deficiency= Cryoprecipitate = just for A.
- Hemophilia B = Factor IX 9 deficiency = Christmas Disease = Fresh frozen plasma = can be use in both A and B

Nota: en el diagnostico las plaquetas están bien.

<u>Assessment</u>

1. Bleeding
    - ✓ Bruising
    - ✓ Subcutaneous hematoma
    - ✓ Epistasis
    - ✓ Gum bleeding
    - ✓ Hematouria
2. Joints bleeding = Hemarthrosis
    - ✓ Pallor
    - ✓ Anesthesia

<u>Intervention</u>

1. **Apply ICE Pack**
2. Avoid contact sport
3. Bleeding medication
4. Handle gently using lift sheet
5. Bleeding Precaution
    - Avoid IM
    - Avoid rectal trauma, enemas
    - Use an electric or safety shaver.
    - Soft- bristled toothbrush, no floss
    - No ASA
    - Stool softener
    - Pad Furniture corner
    - Avoid playing wind instrument
6. Desmopressin or vasopressin
7. Check neurologic status, if any change calls the Dr. (bc brain bleeding, is an emergency)
8. Joint immobilization + elevation + ICE.
9. After any invasive procedure, apply pressure for 15 minutes.
10. Wear protective device = Helmet, knee – elbow pad.

Von Willebrand's Disease

- Hereditary bleeding Disorder
- Autosomal Dominant 50%

<u>Assessment</u>

1. Epistaxis = leaning-forward, pinch nose and apply ICE
2. Gum Bleeding
3. Easily bruising
4. Excessive menstrual bleeding

Intervention

1. Desmopressin or vasopressin (intranasal) are used in diabetes insipid, hemophilia and Von Willebrand's.
2. Bleeding Precaution

B-Thalassemia Major

- Mediterranean Anemia
- Hemolitic Anemia like sicklemia and newborn anemia
- Autosomal recesive 25%.

Assessment

1. Greenish yellow skin tone
2. Micrositic hyperemic anemia (Severe) = frequent blood transfusion, esto conyeba a Hemosiderosis or Iron overload in blood bc red cell rupture.

It is treated with **chelation therapy** = (Deferoxamine) = iron binding (IV) elimina el iron por las feces and causes risk of kidney failure.

3. Frontal bossing
4. Maxillary prominence
5. Wide set, eye with a flattered nose
6. Hepatosplenomegaly

Treatment

1- Blood transfusion
2- Deferoxamine
3- Hemodialysis
4- Splenectomy = Pneumococcal vaccine (risk for infection)
5- Bone Marrow transplant (bed side procedure).

Idiopathic Thrombocytopenic Purpura

- ↓ WBC = Risk for infection
- Platelets less than 150 000
- Risk por spontaneous bleeding less than 50 000
- Platelet's transfusion less than 20 000

Nota: puede convertirse en Hep A

Assessment

1. Small Flat pinpoint red microhemorrhage = petechia
2. Purpura ecchymosis
3. Prolonged bleeding after venipuncture
4. Weakness fatigue, dizziness, tachycardia, abdominal pain, hypotension

Treatment

1. Prednisone (choice)
2. IV Immunoglobulin
3. Steroids
4. Plasmapheresis (filtrar todo el cuerpo) parecido a una hemodiálisis (tto x choice of Guilian Barre)
5. Damasol = decrease immune System
6. Cyclophosphamide = anticancer S/E Cystitis
7. Splenectomy

Cohort:

- Osteomyelitis
- HIV
- Hpt A
- Mononucleosis